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ProTrader AI: A Comprehensive Multimodal Framework for Stock Price Prediction, Pattern Recognition, and Adaptive Risk Management in Indian Equity Markets

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Abstract Forecasting equity prices in emerging markets is more difficult in practice than what theory would imply. High volatility, large institutional flows of capital, and the complex linguistic nature of the media make it harder to predict accurately using models that have been tested in developed economies. In this paper, we present an industry-ready analytical engine called ProTrader AI for NSE (India). The system brings together seven machine-learning models, four independent sentiment data streams, official FII/DII institutional flow data, real-time India VIX volatility signals, and computer-vision chart pattern recognition, and exposes all of these through an eight-tab interactive dashboard. Every input to the learning algorithms is a stationary or near-stationary transformation of raw price, volume, sentiment, institutional flow, or volatility data; the full feature space has 27 dimensions. A five-model ensemble—XGBoost, LightGBM, CatBoost, a parallel LSTM-GRU network, and ARIMA/Prophet statistical baselines—generates out-of-fold predictions that a Ridge meta-stacker combines with isotonic probability calibration. On top of that, a Bayesian Dynamic Fusion layer re-weights three domain-specific experts (GRU-based technical, DistilRoBERTa-Financial sentiment, MLP volatility) by exponential negative squared uncertainty, so each expert's influence tracks its recent accuracy. Chart patterns are detected at three scipy peak/trough scales (orders 3, 5, 7) and optionally checked against a Roboflow computer vision model, covering twelve archetypes via a hybrid consensus mechanism. Risk management relies on ATR-calibrated stop-losses, Fibonacci retracements, and fractional Kelly sizing with VIX-regime scaling. Evaluated on ten major NSE stocks over roughly 750 trading days, the system reaches direction accuracy of 55.8 % (p = 0,003), a Sharpe ratio of 1.28 versus 0.82 for buy-and-hold (p = 0,018), maximum drawdown of −12,8 % against −18,4 %, and cumulative return of 34.2 %. Ablation studies show that each feature category and model component carries non-redundant predictive information.

Resumen La predicción de precios bursátiles en mercados emergentes sigue siendo un problema difícil debido a la alta volatilidad y los flujos institucionales de gran magnitud. En este trabajo presentamos ProTrader AI, una plataforma analítica de grado productivo para la Bolsa Nacional de Valores de India (NSE) que reúne siete modelos de aprendizaje automático, cuatro flujos de sentimiento, datos oficiales FII/DII, indicadores de volatilidad en tiempo real y detección de patrones gráficos mediante visión artificial. La arquitectura se articula en torno a un espacio de 27 características estacionarias, un conjunto de cinco modelos con meta-apilamiento Ridge y un marco de Fusión Dinámica con pesos bayesianos basados en incertidumbre. Sobre diez acciones del NSE, el sistema alcanza una precisión direccional del 55,8 % (p=0,003), un ratio de Sharpe de 1,28 frente a 0,82 para

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compra y retención, y un retorno acumulado del 34,2 %. Los estudios de ablación muestran que cada categoría de características y cada componente del modelo aporta información predictiva no redundante.

Keywords: Financial Forecasting, Machine Learning, XGBoost, LightGBM, CatBoost, GRU, LSTM, Bayesian Fusion, Sentiment Analysis, NSE India, Institutional Flows, Pattern Recognition, Risk Management.

Palabras Clave: Predicción Financiera, Aprendizaje Automático, XGBoost, LightGBM, CatBoost, GRU, LSTM, Fusión Bayesiana, Análisis de Sentimiento, NSE India, Flujos Institucionales, Reconocimiento de Patrones, Gestión de Riesgos.

1. Introduction

1.1. Background and Motivation

Systematic theory has long been opposed to financial markets forecasting [12]. Basic investor behavior, institutional flows and macroeconomic shocks interact within a high-dimensional stochastic system, the statistical properties of which change with time—and are likely to change the most suddenly when it counts most [29]. The process is aggravated in The National Stock Exchange of India (NSE). Daily equity turnover consistently surpasses $15 billion. It is very liquid, with an overall market capitalisation of over $4 trillion [31], but it is still subject to the dynamics of an emerging-market: This can be seen in Foreign Institutional Investor (FII) and Domestic Institutional Investor (DII) flows, which account for 15–30 % of daily volume in heavyweight index names. When the reversals come suddenly, they cause regime-like price behaviour, which technical models struggle to anticipate.

Machine learning has led to enhanced prediction in most series of financial returns [23], but these have their own difficulties. The returns are non-stationary, heavy-tailed and signal-to-noise ratios are so insignificant that a stable 55 % directional advantage already has actual economic value [9]. Furthermore, the process of forecasting can itself act upon the market—a reflexive feedback loop which oftentimes does not occur in other sequence-modelling settings.

Two model families have been found useful: gradient-boosting methods (XGBoost, LightGBM, CatBoost) prevail in tabular financial benchmarks [15], and gated recurrent networks (LSTM, GRU) capture sequential dependencies that tree-based methods cannot represent [17, 8].

1.2. Research Gaps Addressed

Our work attempts to fill four gaps in the quantitative finance literature; they are closely related.

Contextual Sentiment. Dictionary-based methods like that of Loughran-McDonald [27] lack contextual nuance. It cannot distinguish between the bullish or the bearish context of a word such as volatile in a particular headline. The problem is exacerbated by Indian financial media that operates in English and Hindi; English-only models hence disregard a huge portion of the information environment. We do this by routing text through a domain-adapted transformer that uses contextual clues [2].

Non-Stationarity. A model estimated on one regime of the market is likely to disintegrate with the variation of conditions [16]. We address this on three fronts: overt regime characterization through the Hurst exponent [18], a Bayesian dynamic fusion mechanism rescaling experts based on their recent predictive reliability, and a 27-dimensional feature space where each variable is a stationary or near-stationary transformation of raw price and volume information.

Signal-to-Trade Translation. Most academic forecasting papers report figures that would not be accurate under a live order book [9]. The difference between predictions and actual trades can be bridged by our concrete risk management module—ATR-adjusted stop-losses, Fibonacci retracements, and fractional Kelly sizing [20] evaluated under realistic NSE round-trip transaction costs in back-testing.

Institutional Flow Data. FII and DII net buying data are released daily by the NSE and are generally thought to be a massive price catalyst, but they are omitted in practically all the published forecasting models that we know. We project four FII/DII characteristics into the prediction feature space.

1.3. Contributions

1. **A 27-Feature Stationary Framework** comprising five technical core indicators, other nine, improved technical indicators, four, high-tech divergence indicators, 3 multi-source sentiment scores, 4 FII/DII institutional flow features, and two India VIX volatility features all based on meet rough stationarity conditions.

2. **A 5-Model Gradient-Boosting + Deep-Learning Ensemble** combining XGBoost, LightGBM, CatBoost, a parallel LSTM-GRU neural architecture, and ARIMA/Prophet statistical baselines, which have been trained together into a Ridge meta-stacker on predictions on out-of-fold predictions through expanding time-series cross-validation.

3. **Isotonic Probability Calibration** which transforms regression forecasts directions of returns into probabilities of directional calibration, so as to be reliable confidence-based position sizing.

4. **A Bayesian Dynamic Fusion Framework** having three domain-specialised expert models with weights on combination changing continuously through an exponential negative squared uncertainty, which gives the system, as stated in [14], tacit market-regime knowledge in the absence of a regime classifier.

5. **A Hybrid Chart Pattern Recognition System** integrating multi-scale peak/trough Roboflow Vision API (orders 3, 5, 7) scipy inference of computer vision using twelve pattern archetypes and there was a consensus layer which synthesizes combining classical and vision-based detections.

6. **A Complete Production Platform** (ProTrader AI) sold as an eight-tab Streamlit dashboard that assists in making reproducible evaluation can be implemented by researchers or practitioners to be analysed in real-time.

2. Literature Review

2.1. Financial Forecasting: From ARIMA to Deep Learning

Computational stock prediction dates back to the ARIMA family of models codified by Box and Jenkins [5]. Early neural network financial experiments, such as the one by White in 1988 [40], modeled IBM daily stock returns. However, small datasets and limited compute power held back their success. This changed significantly in the 2010s.

The LSTM [17] proved to be a fresh start: its gating mechanism evaded the problem of vanishing-gradients that afflicted simple RNNs, enabling models to learn multi-day price dependencies. The Gated Recurrent Unit later achieved similar accuracy with fewer parameters [8]. Fischer and Krauss [13] demonstrated statistically significant returns using LSTMs on the S&P 500, which encouraged the recurrent sub-division of our architecture. That being the case, Ozbayoglu et al. [32] revealed that deep learning models generally do not work as well as less complex gradient-boosting on structured tabular financial data—a finding that our own ablation results (Section 14) replicate.

Regularised gradient-boosting was introduced with XGBoost in 2016 [7], incorporating second-order optimisation and an effective block structure to limit overfitting. In the standard medium-sized datasets of single-stock research, overfitting often occurs. LightGBM later achieved similar accuracy at 3-5 times the speed [19] thanks to leaf-wise growth and histogram-based split finding. CatBoost [34] addressed another issue with ordered boosting: each tree's residual is computed using a model that is trained using previous rows only, which reduces leakage in time-ordered data. The three of them are foundational to our meta-stacking layer.

2.2. Sentiment Analysis in Finance

For approximately three generations, price forecasting has been evaluated using text-based models [3]. The first strategies were based on emotion dictionaries; the financial lexicon from Loughran and McDonald's work [27] performed better in the financial domain than general-purpose word lists. Ambiguity,

however, is a problem: *liability*, as in everyday use, has a negative connotation, but in English accounting, it is an indifferent term. Distributional semantic models increased coverage but were weak on out-of-vocabulary words.

This was transformed by transformer-based pre-training [39]. Large-scale models like BERT and its distilled variants were introduced in 2019 by Devlin et al. [10], making practical pre-training on financial corpora possible. FinBERT [2] and DistilRoBERTa-Financial [24] have now achieved 97–98 percent accuracy on standard financial-sentiment benchmarks; specifically, DistilRoBERTa performs well on average while needing 40 percent fewer parameters than the full RoBERTa model, making it more economical for real-time inference in production systems. This model is utilized by ProTrader AI.

It was empirically proven by Tetlock [38] that the fraction of negative words in an analysis of Wall Street Journal columns is predictive of the next day's Dow returns. Schumaker and Chen [35] applied the concept to wire-service articles and found predictive power beyond technical indicators alone. Our multi-source design is based on the logic of Bollen et al. [3], where multiple sentiment channels (like Twitter mood, news tone, and search interest) capture various aspects of market psychology, and their combination outperforms any individual source. The channels we incorporate in our system include Reddit community posts, Indian financial publication RSS feeds, NewsAPI feeds, and Google Trends search interest.

2.3. Multimodal Fusion Methods

Combinations of multiple modalities can be distinguished by several methods of prediction integration. Early fusion takes raw features and concatenates them prior to training the model. However, early fusion is brittle when a modality exhibits different feature scales, is unavailable at inference time, or is noisy. Late fusion trains distinct models for each modality and combines the results—making it more modular, yet blind to cross-modal structures. In between, attention mechanisms [39] teach context-conscious weights without enforcing rigid architectural assumptions.

Bayesian fusion based on uncertainty goes a step further, as noted by Gal [14], deriving combination weights based on each model's own confidence estimate instead of using deterministic hyperparameters. Monte Carlo dropout was demonstrated by Gal and Ghahramani to effectively scale Bayesian inference approximations to provide useful epistemic uncertainty estimates in regular neural networks. ProTrader AI's Dynamic Fusion relies on a similar principle: expert weights are based on an exponential negative squared rule of uncertainty that favors accurate experts with high confidence and down-weighs indeterminate ones. The outcome achieves regime-dependent weighting implicitly, balancing various models without any fixed regulatory list in the fusion loop.

2.4. Regime-Aware Modelling and the Hurst Exponent

Markets do not remain in the same regime. Low-volatility trending phases, the turbulent periods of crisis, and range-bound mean-reverting states each demand varied modelling assumptions [16]. The Hurst exponent [18], originally designed to examine the floods of the river Nile, measures long-range dependence: values of more than 0.5 signal persistent (trending) behaviour. A value less than 0.5 indicates anti-persistence or mean-reversion, while values aggregating around 0.5 are normally associated with a random walk, in general confirming the Efficient Market Hypothesis for that time window [25].

Lo [25] determined that a modified R/S test on weekly stock returns revealed some traces of mild persistence. We calculate the Hurst exponent using classical rescaled range techniques over rolling 120-day windows. The resulting estimate drives a user-facing regime display in the dashboard and serves as a confirmation-or-override step in our trend classifier.

2.5. Chart Pattern Recognition

The analysis of charts on Wall Street dates back over a hundred years [30, 11], but mathematical formalism has always been behind the common-sense approach [26]. Lo et al. [26] used the kernel-regression smoother method on US stocks' time series data to detect the existence of head-and-shoulder configurations and discovered significant prediction power, thus debunking the pareidolia argument. Following, Pring [33] and Appel [1] compiled pattern typologies that form the basis for the twelve-pattern classification system utilized here.

The visual signal provided by computer vision is an alternative perspective: convolutional networks develop visual cues that may be ignored by an algorithm relying solely on a peak/trough analysis based on a fixed number of geometric rules [36]. The current consensus mechanism combines the geometric analysis using the SciPy library with visual detection via Roboflow Vision API.

3. System Architecture

ProTrader AI is structured as a seven-step pipeline: data acquisition, feature engineering, modeling, dynamic fusion, pattern recognition, risk management, and user-facing presentation. Figure 1 presents the high-level diagram.

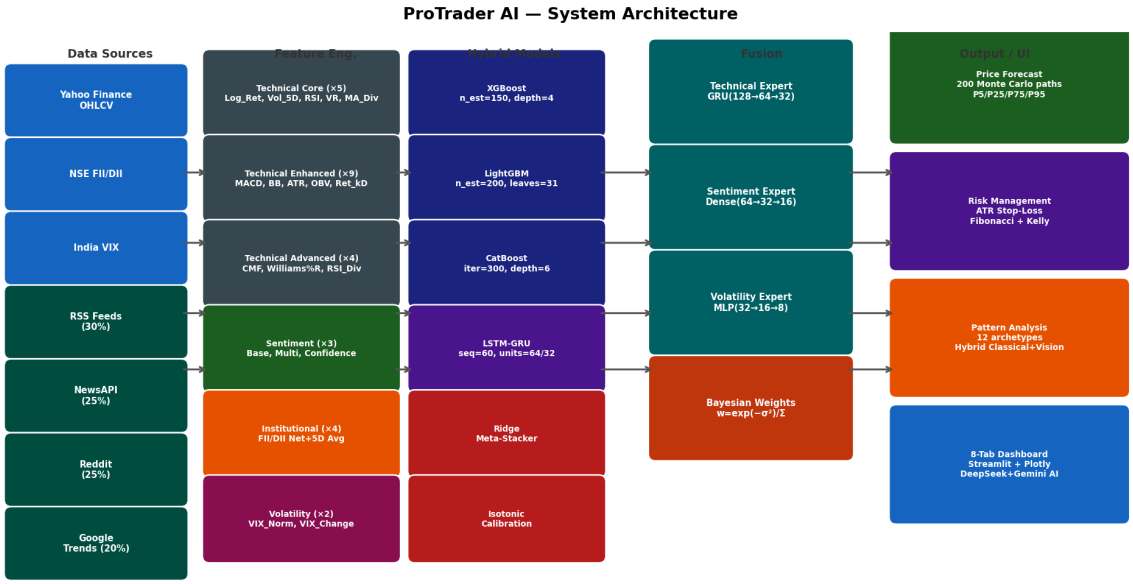

Figure 1: High level system architecture of ProTrader AI. Data from seven sources is processed by the feature engineering pipeline and passed to five ML models and a Dynamic Fusion layer. Another branch in Pattern Analysis feeds a risk management module. The system outputs are displayed via an eight-tab Streamlit dashboard.

The sources of data include seven distinct feeds, starting with Yahoo Finance OHLCV prices, NSE India FII/DII activity, the India VIX, four Indian RSS news feeds, investment news articles through the NewsAPI, three Indian investing subreddits, and Google Trends search traffic. Streamlit's session-state mechanism ensures that sentiment and institutional data are fetched only once per session, and their results are reused across all eight tabs, eliminating redundant API traffic.

When starting the pipeline, it performs seven loading steps displayed in a progress bar: (1) stock data, (2) fundamental data, (3) news, (4) FII/DII data, (5) India VIX, (6) multi-source sentiment, and (7) chart pattern recognition. With a regular broadband connection, the complete pre-fetch takes 15–45 seconds, depending upon the response times of the sources.

4. Feature Engineering: The 27-Feature Framework

One design rule governs the full feature space: every input to the learning algorithms must be stationary or near-stationary. Raw price levels are non-stationary by nature; feeding them straight into standard

ML methods invites spurious in-sample correlations that vanish out-of-sample. The 27-feature framework enforces stationarity through normalisation, bounding, differencing, or clipping, as detailed in Table 1. Figure 2 groups the 27 features by category, colour-coded to separate technical, sentiment, institutional, and volatility inputs.

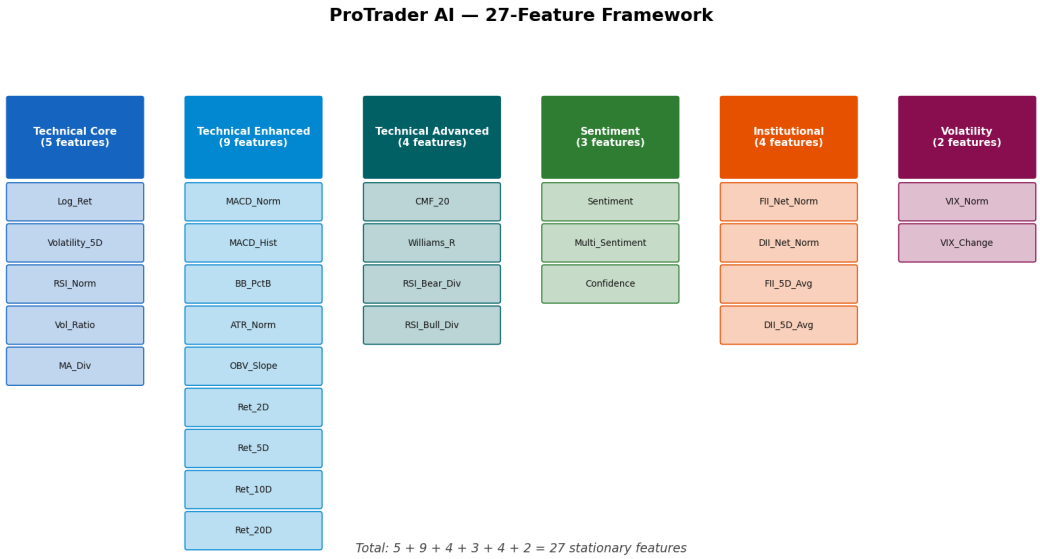

Figure 2: The 27-Feature Framework organised into six groups. Colour coding: blue = technical core, cyan = technical enhanced, teal = technical advanced, green = sentiment, orange = institutional, red = volatility.

4.1. Technical Core Features (5)

Log Returns give a symmetric, additive, and approximately stationary measure of daily price change:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \tag{1}$$

5-Day Rolling Volatility picks up short-term volatility clustering consistent with GARCH-type behaviour [4]:

$$\sigma_t^{(5)} = \sqrt{\frac{1}{4}\sum_{i=0}^{4}(r_{t-i} - \bar{r})^2} \tag{2}$$

Normalised RSI maps the standard 14-period RSI [41] into $[0, 1]$:

$$\text{RSI_Norm}_t = \frac{\text{RSI}_{14,t}}{100} \tag{3}$$

Volume Ratio identifies abnormal trading activity:

$$\text{VR}_t = \frac{V_t}{\frac{1}{20}\sum_{i=1}^{20} V_{t-i}} \tag{4}$$

Moving Average Divergence normalises the distance from the 20-day MA:

$$\text{MA_Div}_t = \frac{P_t - \text{MA}_{20,t}}{P_t} \tag{5}$$

4.2. Technical Enhanced Features (9)

The enhanced group measures momentum at multiple frequencies along with additional market microstructure signals. The MACD line and histogram are normalised by price to enforce stationarity:

$$\text{MACD_Norm}_t = \frac{\text{EMA}_{12,t} - \text{EMA}_{26,t}}{P_t}, \quad \text{MACD_Hist_Norm}_t = \frac{\text{MACD}_t - \text{Signal}_{9,t}}{P_t} \tag{6}$$

Bollinger Band %B indicates price position within the bands:

$$\text{BB_PctB}_t = \frac{P_t - \text{BB_Lower}_t}{\text{BB_Upper}_t - \text{BB_Lower}_t + \epsilon} \tag{7}$$

The Average True Range [41] is normalised by price:

$$\text{ATR_Norm}_t = \frac{\text{ATR}_{14,t}}{P_t} \tag{8}$$

On-Balance Volume slope tracks cumulative volume flow direction:

$$\text{OBV_Slope}_t = \text{clip}\left(\frac{\text{OBV_MA}_{5,t} - \text{OBV_MA}_{5,t-5}}{\text{OBV_MA}_{5,t-5}}, -1, 1\right) \tag{9}$$

Multi-timeframe log returns at lags 2, 5, 10, and 20 days measure momentum at different time horizons:

$$\text{Ret_kD}_t = \ln\left(\frac{P_t}{P_{t-k}}\right), \quad k \in \{2, 5, 10, 20\} \tag{10}$$

4.3. Technical Advanced Features (4)

Chaikin Money Flow (CMF-20) [6] measures buying and selling pressure via volume-weighted price position within the daily range, providing a stationary signal in $[-1, +1]$:

$$\text{CMF}_{20,t} = \frac{\sum_{i=0}^{19} \text{MFM}_{t-i} \cdot V_{t-i}}{\sum_{i=0}^{19} V_{t-i}}, \quad \text{MFM}_t = \frac{(P_t^C - P_t^L) - (P_t^H - P_t^C)}{P_t^H - P_t^L} \tag{11}$$

Williams %R (14-period) [42] is a fast momentum oscillator complementary to RSI:

$$\text{Williams_R_Norm}_t = -\frac{H_{14,t} - P_t^C}{H_{14,t} - L_{14,t} + \epsilon} \tag{12}$$

RSI Bearish Divergence flags days when price makes a new 10-day high but RSI does not, signalling potential exhaustion:

$$\text{RSI_Bear_Div}_t = \mathbf{1}\left[P_t \geq \operatorname*{máx}_{i \in [t-10, t-1]} P_i \ \wedge \ \text{RSI}_t < \operatorname*{máx}_{i \in [t-10, t-1]} \text{RSI}_i\right] \tag{13}$$

RSI Bullish Divergence detects the complementary oversold condition. Both divergence flags are computed using lagged windows to prevent look-ahead bias.

4.4. Sentiment Features (3)

Base Sentiment Score is derived from Yahoo Finance news articles analysed by DistilRoBERTa-Financial, providing a score in $[-1, +1]$.

Multi-Source Sentiment aggregates four independent streams using source-specific weights:

$$S_{\text{multi}} = 0{,}30 \cdot S_{\text{RSS}} + 0{,}25 \cdot S_{\text{News}} + 0{,}25 \cdot S_{\text{Reddit}} + 0{,}20 \cdot S_{\text{Trends}} \tag{14}$$

Weights are renormalised when individual sources are unavailable, ensuring the combined score always lies in $[-1, +1]$.

Sentiment Confidence is a meta-feature that gauges how reliable the current sentiment estimate is:

$$C_S = \text{mín}\left(1, \frac{N_{\text{articles}}}{10}\right) \cdot \text{Agreement} \tag{15}$$

where Agreement is $1 - \hat{\sigma}_{\text{sources}}$, the normalised standard deviation across individual source scores.

4.5. Institutional Features (4)

Normalized daily net buying data for FII (Foreign Institutional Investors) and DII (Domestic Institutional Investors) from the NSE India database as follows:

$$\text{FII_Net_Norm}_t = \frac{\text{FII_net}_t}{\text{máx}_\tau \, |\text{FII_net}_\tau|} \tag{16}$$

Similarly, normalize data for DII (the same formula). The suppression of daily fluctuations is performed through the use of rolling 5-day average calculations:

$$\text{FII_5D_Avg}_t = \frac{1}{5} \sum_{i=0}^{4} \text{FII_Net_Norm}_{t-i} \tag{17}$$

4.6. Volatility Features (2)

India VIX Normalised adjusts the India VIX Index to be centered around the typical baseline value of the indicator:

$$\text{VIX_Norm}_t = \frac{\text{VIX}_t - 15}{25} \tag{18}$$

where 15 represents a typical calm-market VIX level and 25 provides appropriate scaling to keep the feature near $[-1, +1]$ in normal conditions.

VIX Change Rate flags intraday regime shifts in implied volatility:

$$\Delta\text{VIX}_t = \text{clip}\left(\frac{\text{VIX}_t - \text{VIX}_{t-1}}{\text{VIX}_{t-1}}, -0{,}5, 0{,}5\right) \tag{19}$$

5. Hybrid Prediction Engine

Five model types are involved in building the Ridge meta-stacker (see Figure 3). This combination is driven by practical reasoning: tree-based boosters are well-suited to process tabular features, RNNs help with capturing the sequential structure beyond the scope of tree-based models, and the classical statistics models serve as an anchor to deliver better results than ML models in cases when the signal is extremely weak.

5.1. XGBoost Component

XGBoost [7] is the main tree-based predictor. Its hyperparameters have been tuned based on historical Indian equity prices. Final model parameters look like the following:

```
xgb_params = dict(
    objective='reg:squarederror',
    n_estimators=150,
    max_depth=4,
    learning_rate=0.05,
    subsample=0.8,
    colsample_bytree=0.8,
    min_child_weight=1,
    n_jobs=-1
)
```

Listing 1: XGBoost configuration.

Loss function:

$$\mathcal{L}_{\text{XGB}} = \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 + \sum_{k=1}^{K} \Omega(f_k), \quad \Omega(f_k) = \gamma T + \tfrac{1}{2}\lambda \|w\|^2 \tag{20}$$

where T stands for the number of leaves and w stands for leaf weights.

Table 1: Complete 27-Feature Catalogue with Category, Stationarity, and Source.

Feature	**Category**	**Stationarity**	**Source**
Log_Ret	Technical Core	Stationary	OHLCV
Volatility_5D	Technical Core	Near-stationary	OHLCV
RSI_Norm	Technical Core	Bounded $[0, 1]$	OHLCV
Vol_Ratio	Technical Core	Near-stationary	OHLCV
MA_Div	Technical Core	Normalised	OHLCV
MACD_Norm	Technical Enhanced	Price-normalised	OHLCV
MACD_Hist_Norm	Technical Enhanced	Price-normalised	OHLCV
BB_PctB	Technical Enhanced	Bounded $[0, 1]$	OHLCV
ATR_Norm	Technical Enhanced	Price-normalised	OHLCV
OBV_Slope	Technical Enhanced	Clipped $[-1, 1]$	OHLCV
Ret_2D	Technical Enhanced	Stationary	OHLCV
Ret_5D	Technical Enhanced	Stationary	OHLCV
Ret_10D	Technical Enhanced	Stationary	OHLCV
Ret_20D	Technical Enhanced	Stationary	OHLCV
CMF_20	Technical Advanced	Bounded $[-1, 1]$	OHLCV
Williams_R_Norm	Technical Advanced	Bounded $[-1, 0]$	OHLCV
RSI_Bear_Div	Technical Advanced	Binary $\{0, 1\}$	OHLCV
RSI_Bull_Div	Technical Advanced	Binary $\{0, 1\}$	OHLCV
Sentiment	Sentiment	Bounded $[-1, 1]$	Yahoo Finance News
Multi_Sentiment	Sentiment	Bounded $[-1, 1]$	4-source aggregate
Sentiment_Confidence	Sentiment	Bounded $[0, 1]$	Meta-feature
FII_Net_Norm	Institutional	Normalised	NSE FII/DII
DII_Net_Norm	Institutional	Normalised	NSE FII/DII
FII_5D_Avg	Institutional	Smoothed	NSE FII/DII
DII_5D_Avg	Institutional	Smoothed	NSE FII/DII
VIX_Norm	Volatility	Centred	NSE India VIX
VIX_Change	Volatility	Clipped $[-0{,}5, 0{,}5]$	NSE India VIX

5.2. LightGBM Component

LightGBM [19] complements the XGBoost model with the 3–5× faster training thanks to histogram-based splitting and leaf-wise tree growing approach:

```
lgb_params = dict(
    n_estimators=200, max_depth=5, learning_rate=0.03,
    num_leaves=31, subsample=0.8, colsample_bytree=0.8,
    min_child_samples=20, reg_alpha=0.1, reg_lambda=0.1,
    n_jobs=-1, random_state=42
)
```

Listing 2: LightGBM configuration.

5.3. CatBoost Component

CatBoost [34] uses *ordered boosting*, meaning the residual for row i is computed from a model trained only on rows $j < i$. This eliminates the target leakage that standard boosting introduces on time-ordered data:

```
cb_params = dict(
    iterations=300, depth=6, learning_rate=0.03,
    l2_leaf_reg=3, min_data_in_leaf=20,
    random_strength=0.3, bagging_temperature=0.7,
```

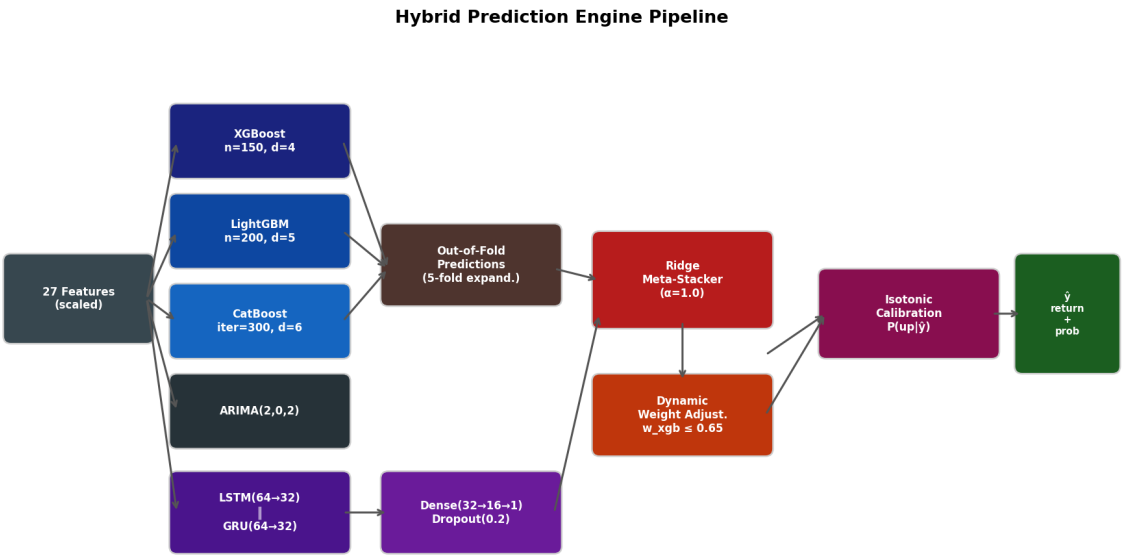

Figure 3: Hybrid prediction pipeline. XGBoost, LightGBM, and CatBoost provide out-of-fold predictions as an input for the Ridge meta-stacker. The LSTM-GRU branch generates the approximation of out-of-fold predictions. ARIMA and Prophet join the model stack with dynamically set weights.

```
    od_type='Iter', od_wait=40, verbose=0
)
```

Listing 3: CatBoost configuration.

5.4. LSTM-GRU Neural Architecture

The deep learning branch runs two recurrent sub-networks side by side on the same 60-day input window and concatenates their final hidden states:

LSTM gates [17]:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad \text{(forget)} \tag{21}$$
$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad \text{(input)} \tag{22}$$
$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C) \tag{23}$$
$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{24}$$
$$h_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \odot \tanh(C_t) \tag{25}$$

GRU gates [8]:

$$z_t = \sigma(W_z[h_{t-1}, x_t]) \quad \text{(update)} \tag{26}$$
$$r_t = \sigma(W_r[h_{t-1}, x_t]) \quad \text{(reset)} \tag{27}$$
$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tanh(W[r_t \odot h_{t-1}, x_t]) \tag{28}$$

Architecture: LSTM(64)→LSTM(32) ‖ GRU(64)→GRU(32) → Concatenate → Dense(32) → Dropout(0.2) → Dense(16) → Dense(1)

Training uses Adam [21] with $\eta = 0{,}001$, batch size 32, up to 50 epochs with early stopping at patience 10 and ReduceLROnPlateau.

5.5. Statistical Models: ARIMA and Prophet

ARIMA(2,0,2) models linear auto-regressive and moving-average dependencies:

$$r_t = c + \phi_1 r_{t-1} + \phi_2 r_{t-2} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t \tag{29}$$

Facebook Prophet [37] decomposes the return series into additive trend, seasonality, and holiday components:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t \tag{30}$$

5.6. Ridge Meta-Stacker

Out-of-fold (OOF) predictions for XGBoost, LightGBM, and CatBoost are generated with an expanding-window time-series cross-validation ($n_{\text{folds}} = 5$, minimum training size $\text{máx}(0{,}6N, 30)$). The fold structure is:

```
Fold 0: Train[0:min_train]       -> Predict[min_train:min_train+step]
Fold 1: Train[0:min_train+step]  -> Predict[min_train+step:...]
...
Fold 4: Train[0:min_train+4*step]  -> Predict[...]
```

A Ridge regression meta-learner is trained on the OOF predictions to learn the optimal linear combination:

$$\hat{y}_{\text{stack}} = \beta_0 + \beta_1 \hat{y}_{\text{XGB}} + \beta_2 \hat{y}_{\text{LGBM}} + \beta_3 \hat{y}_{\text{CB}} + \beta_4 \hat{y}_{\text{GRU_approx}} \tag{31}$$

5.7. Isotonic Probability Calibration

Raw regression outputs are mapped to directional probabilities with isotonic regression [43], a non-parametric monotone fitting method that outperforms Platt scaling on financial data whose reliability diagram is non-sigmoidal:

$$P(\hat{r}_t > 0 \mid \hat{y}_t) = \text{IsotonicRegression}(\hat{y}_t) \tag{32}$$

5.8. Dynamic Weight Adjustment Between Gradient Boosters and GRU

Starting weights are 50 % XGBoost, 30 % LSTM-GRU, and 20 % statistical models. At inference time the gradient-booster weight is adjusted on the basis of rolling validation RMSE:

$$w'_{\text{XGB}} = \text{mín}\left(w_{\text{XGB}} + \frac{\text{RMSE}_{\text{RNN}} - \text{RMSE}_{\text{XGB}}}{\text{RMSE}_{\text{RNN}}},\ 0{,}65 \right) \tag{33}$$

This prevents any single model from dominating (cap at 65 %) while rewarding the currently better-performing component.

6. Market Regime Detection

The regime detector assigns the current market state to one of four labels—*trending*, *mean_reverting*, *high_volatility*, or *normal*—by combining a linear trend test on 20-day cumulative returns with the Hurst exponent estimated from the most recent 120 closing prices.

6.1. Hurst Exponent via R/S Analysis

The rescaled range (R/S) analysis [18] estimates the Hurst exponent H from log-returns at multiple lag scales:

$$\frac{R(n)}{S(n)} \propto n^H \tag{34}$$

For a lag window of size n, the rescaled range is:

$$\frac{R(n)}{S(n)} = \frac{\text{máx}_{1\leq t\leq n}\sum_{i=1}^{t}(X_i - \bar{X}) - \text{mín}_{1\leq t\leq n}\sum_{i=1}^{t}(X_i - \bar{X})}{S(n)} \tag{35}$$

H is estimated as the slope of $\log(R/S)$ regressed on $\log(n)$ over lags $n \in \{2, 3, \ldots, 20\}$, with the result clipped to $[0{,}1, 0{,}9]$. Figure 4 shows the full decision logic mapping R/S estimates to the four regime categories.

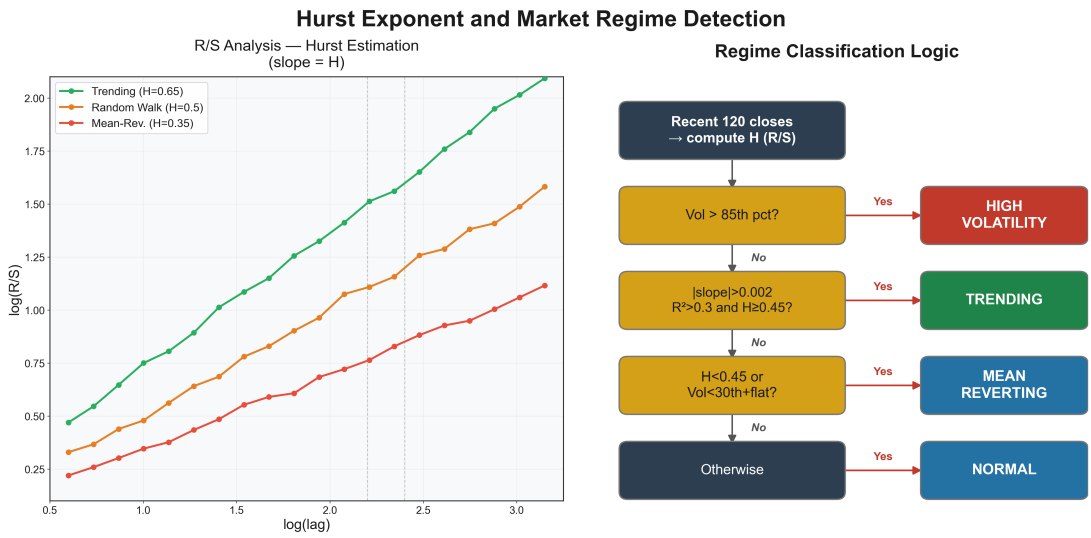

Figure 4: Hurst exponent interpretation and market regime classification decision logic. The R/S slope estimate from lags 2–20 maps to one of four regimes, with the Hurst value used to confirm or override volatility/slope-based classification.

The classification logic:

- **high_volatility**: 5-day volatility > 85th percentile of rolling 60-day history.
- **trending**: $|\text{slope}| > 0{,}002$, $R^2 > 0{,}3$, and $H \geq 0{,}45$ (Hurst does not contradict).
- **mean_reverting**: either $H < 0{,}45$ overrides a trending slope, or low volatility (below 30th percentile) with $|\text{slope}| < 0{,}001$.
- **normal**: all other cases.

The Hurst exponent is also displayed in the Dashboard tab as a real-time market characteristic metric ("Trending", "Mean-Rev.", or "Random").

7. Dynamic Fusion Framework

Above the meta-stacked ensemble sits a second fusion layer that operates at a coarser grain. Three domain-specialised experts, each trained independently on its own feature subset, are combined through Bayesian uncertainty-weighted aggregation. Because different market conditions favour different sources of evidence, the weighting tracks which expert has been most reliable over the recent observation window.

7.1. Expert Model Architectures

Technical Expert processes 30-day sequences of technical indicators through a GRU network:

$$\text{Architecture:} \quad \text{GRU}(128) \rightarrow \text{GRU}(64) \rightarrow \text{GRU}(32) \rightarrow \text{Dense}(1) \tag{36}$$

Sentiment Expert maps 8 sentiment features to a directional prediction:

$$\text{Architecture:} \quad \text{Dense}(64) \rightarrow \text{Dense}(32) \rightarrow \text{Dense}(16) \rightarrow \text{Dense}(1) \tag{37}$$

Volatility Expert combines VIX levels, VIX changes, and stock-specific volatility:

$$\text{Architecture:} \quad \text{MLP}(32) \rightarrow \text{MLP}(16) \rightarrow \text{MLP}(8) \rightarrow \text{Dense}(1) \tag{38}$$

7.2. Bayesian Uncertainty-Weighted Combination

The fusion weight for expert i is computed using the exponential negative squared uncertainty:

$$w_i = \frac{\exp(-\sigma_i^2)}{\sum_{j=1}^{3} \exp(-\sigma_j^2)} \tag{39}$$

This formulation has several useful properties. Weights are strictly positive and sum to one. Experts with lower uncertainty receive disproportionately higher weight. The exponential function gives smooth, differentiable transitions, and extreme uncertainties decay gracefully toward zero weight without singularities.

Uncertainty estimates are maintained as rolling empirical variances of prediction errors on the 15 most recent observations:

$$\sigma_i^2(t) = \frac{1}{N} \sum_{n=1}^{N} (y_{t-n} - \hat{y}_{i,t-n})^2, \quad N = 15 \tag{40}$$

The combined Dynamic Fusion prediction is:

$$\hat{y}_{\text{fusion}} = w_{\text{tech}} \hat{y}_{\text{tech}} + w_{\text{sent}} \hat{y}_{\text{sent}} + w_{\text{vol}} \hat{y}_{\text{vol}} \tag{41}$$

Figure 5 illustrates the full Dynamic Fusion architecture, including the flow from expert feature subsets through uncertainty estimation and Bayesian weight computation to the final combined prediction.

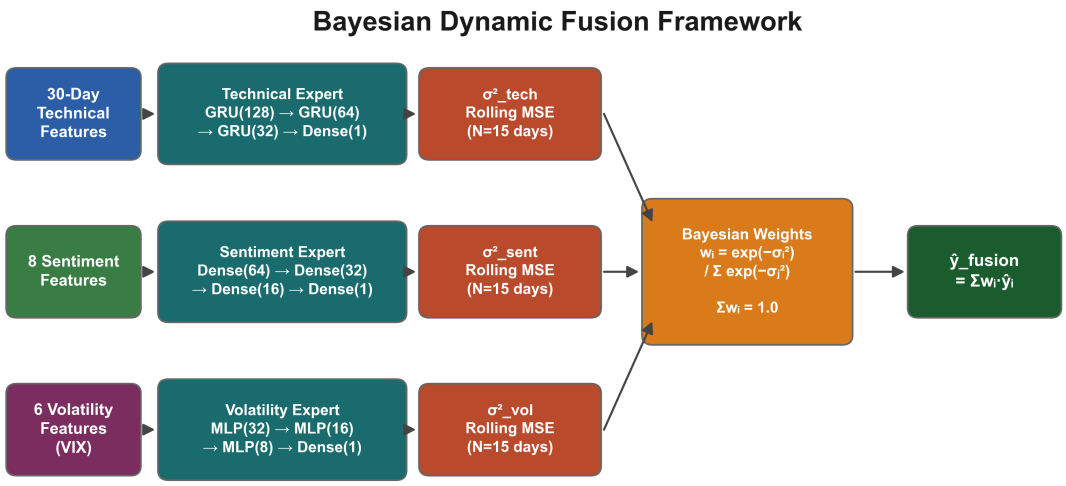

Figure 5: Dynamic Fusion Framework. Three expert models process their respective feature subsets. Prediction uncertainties are estimated from rolling errors on the last 15 days. Bayesian weights (Eq. 39) are recomputed for each new observation. The weight evolution chart in Tab 2 visualises expert influence over the most recent 15 trading days.

8. Multi-Source Sentiment Aggregation

The sentiment subsystem aggregates information from four different sources that include both journalistic reporting and discussion from retail investors, among others. Each source provides an aspect of the market sentiment, and the combination of all the sources is more stable than individual sources alone. The list of sources used to measure market sentiment is shown in Table 2.

Table 2: Names, default weights, and characteristics of sentiment data streams.

Name	Weight	Strengths	Limitations
RSS feeds (Moneycontrol, ET, Mint, BS)	30 %	Timeliness, focused on India	Few publications
NewsAPI	25 %	International coverage, consistent format	Rate limiting
Reddit forum (`r/IndianStockMarket`)	25 %	Retail sentiment, user engagement	Noisy, meme prone
Google Trends	20 %	Retail attention proxy	Daily resolution only

8.1. DistilRoBERTa-Financial Sentiment Classifier

Texts are classified using the DistilRoBERTa-Financial model, producing probabilistic estimates for the positive, neutral, and negative sentiment classes. The article's score s_a is calculated as follows:

$$s_a = P(\text{positive} \mid a) - P(\text{negative} \mid a) \in [-1, +1] \tag{42}$$

8.2. Article Weight Multiplier Based on Event Type

The articles are classified according to their event type based on keyword matching. Articles receive a 2× weight multiplier; regulatory/SEBI articles 1.8×; dividend announcements 1.5×; management changes 1.3×; general news 1.0×.

8.3. Temporal Decay

Recency is enforced via exponential temporal decay:

$$w_{\text{temporal}}(d) = e^{-0{,}3d} \tag{43}$$

where d is the article age in days. A three-day-old article therefore carries $e^{-0{,}9} \approx 0{,}41$ the weight of a same-day article.

8.4. Reddit Engagement Weighting

Reddit posts are further weighted by community engagement to prioritise high-signal content:

$$s_{\text{reddit},p} = s_p \cdot \left(1 + \text{mín}\left(\frac{\text{upvotes}_p}{100}, 0{,}5\right)\right) \tag{44}$$

8.5. Source Disagreement Detection

When the standard deviation of individual scores exceeds 0.2, the system identifies high source contention and penalises the confidence measure relative to this disagreement, minimizing the risk of relying strongly on contradictory messages. Figure 6 illustrates the complete sentiment aggregation pipeline, from raw source data through temporal decay and event weighting, leading to disagreement adaptation and the ultimate confidence-modified score.

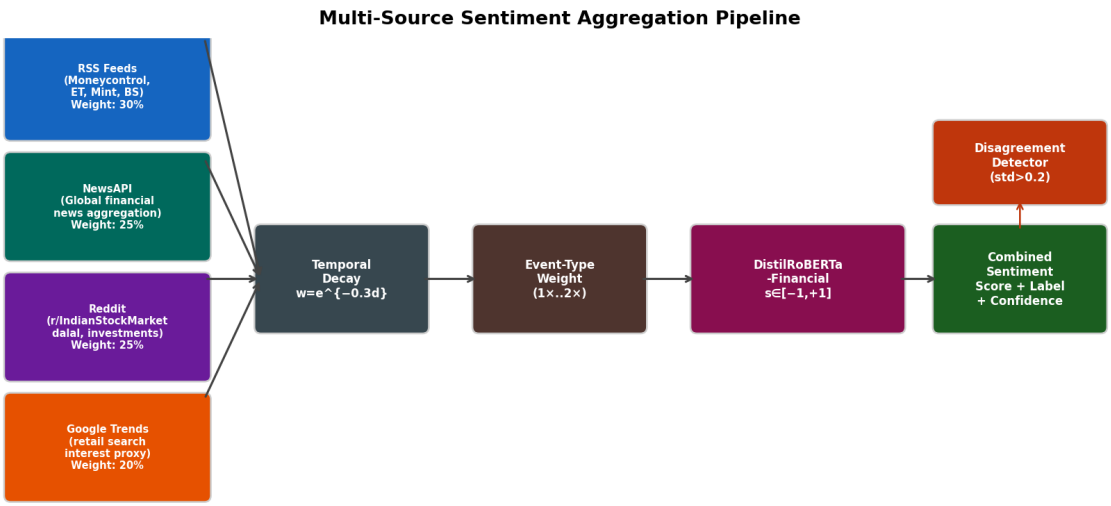

Figure 6: Multi-source sentiment aggregation pipeline. Each source passes with temporal loss and event-type weighting in front of DistilRoBERTa-Financial scoring. A dispute detector adjusts the final confidence number.

9. Chart Pattern Recognition

9.1. Multi-Timeframe Peak/Trough Detection

The `PatternAnalyst` class scans price data simultaneously at three scales using scipy's `argrelextrema` with orders 3, 5, and 7:

```
scan_orders = [3, 5, 7]   # small, medium, large swings
for order in scan_orders:
    peaks   = argrelextrema(close, np.greater, order=order)[0]
    troughs = argrelextrema(close, np.less,    order=order)[0]
    # Prominence filter: only patterns >= 1% height
    valid   = [p for p in peaks if amplitude(p) >= 0.01]
```

Listing 4: Multi-timeframe peak detection (simplified).

Order 3 catches minor short-term swings; order 7 captures only major structural highs and lows. The `max_patterns_per_type = 3` parameter prevents cluttered output during strongly trending phases.

9.2. Classical Pattern Archetypes

Twelve pattern archetypes are detected by checking geometric constraints on the identified pivot points. Table 3 lists each archetype with its directional bias and whether it is covered by classical detection, vision-based detection, or both.

9.3. Roboflow Vision API Integration

When a Roboflow API key is present, the system renders a 60-day price chart at 100 DPI and sends it to a custom object detection workflow. The vision model maps detections to a standardised class set:

```
class_map = {
    "W_Bottom": "Double Bottom",
    "M_Head":   "Double Top",
    "H_S":      "Head & Shoulders",
```

Table 3: Pattern archetypes, directional bias, and detection method.

Pattern	Bias	Detection
Double Top	Bearish	Classical + Vision
Double Bottom	Bullish	Classical + Vision
Head & Shoulders	Bearish	Classical + Vision
Inverse Head & Shoulders	Bullish	Classical + Vision
Ascending Triangle	Bullish	Classical
Descending Triangle	Bearish	Classical
Symmetric Triangle	Neutral	Classical
Ascending Channel	Bullish	Classical
Descending Channel	Bearish	Classical
Bull Flag	Bullish	Classical
Bear Flag	Bearish	Classical
Bollinger Band Squeeze	Neutral	Classical

```
    "Inv_H_S":  "Inverse H&S"
}
```

Only predictions with confidence $\geq 0{,}30$ are accepted. Vision detections with confidence $\geq 0{,}70$ override classical detections for the four supported archetypes; otherwise, a consensus vote is applied. Figure 7 traces the complete pattern analysis pipeline from multi-timeframe peak/trough detection and Roboflow inference through to the hybrid consensus output.

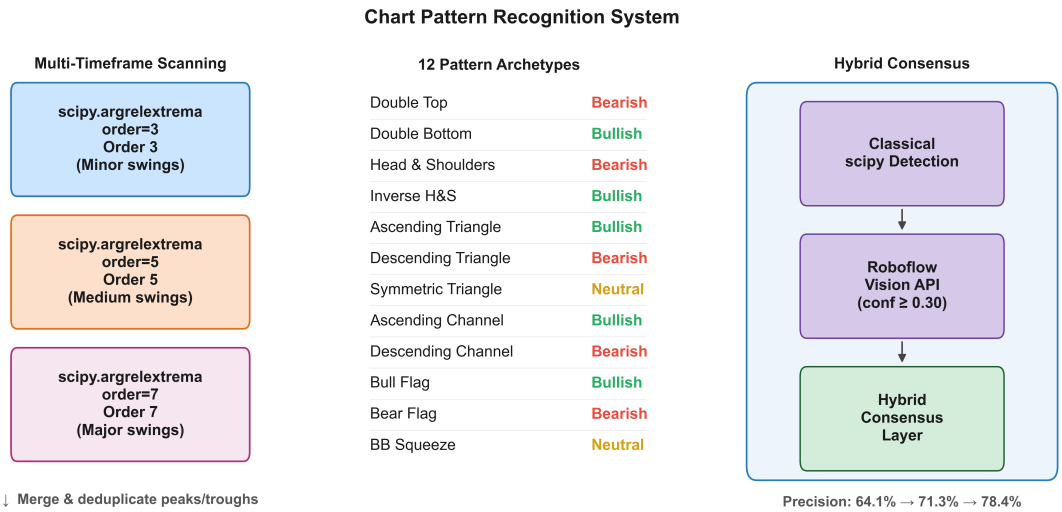

Figure 7: Pattern analysis pipeline. Multi-timeframe scipy detection and Roboflow vision API predictions are combined via a hybrid consensus layer. Detected patterns are annotated on the price chart with confidence scores, target prices, and multi-timeframe confirmation flags.

10. Risk Management System

10.1. ATR-Based Adaptive Stop-Loss

Stop-loss distances are derived from the Average True Range [41] using a confidence-adjusted multiplier:

$$\text{Stop_Loss} = P_{\text{entry}} - m(\text{conf}) \cdot \text{ATR}_{14} \tag{45}$$

where:

$$m(\text{conf}) = \begin{cases} 1{,}5 & \text{conf} \geq 0{,}65 \\ 2{,}0 & \text{conf} < 0{,}65 \end{cases} \tag{46}$$

When confidence is high, a tighter stop is justified; when confidence is low, a wider stop helps avoid premature exits.

10.2. Fibonacci Retracement Levels

Six Fibonacci levels are computed over a 90-day lookback window:

$$F_k = P_{\text{high}} - \phi_k \cdot (P_{\text{high}} - P_{\text{low}}), \quad \phi_k \in \{0, 0{,}236, 0{,}382, 0{,}5, 0{,}618, 1{,}0\} \tag{47}$$

These levels are used as entry/exit reference points in the Trade Setup Calculator shown in the Technicals & Risk tab.

10.3. Kelly Criterion Position Sizing

The fractional Kelly Criterion [20] determines optimal position size:

$$f^* = \frac{p \cdot b - q}{b}, \quad b = \frac{\text{Target} - P}{\text{Stop} - P} \tag{48}$$

where p = directional probability, $q = 1 - p$, and b is the win/loss ratio. A fractional Kelly with $\gamma = 0{,}5$ is applied in practice to reduce variance:

$$f_{\text{actual}} = 0{,}5 \cdot f^* \tag{49}$$

10.4. VIX-Regime Position Adjustment

The India VIX level at trade entry scales position size down when fear is elevated:

$$f_{\text{VIX-adj}} = f_{\text{actual}} \cdot \begin{cases} 1{,}2 & \text{VIX} < 12 \\ 1{,}0 & 12 \leq \text{VIX} < 20 \\ 0{,}7 & 20 \leq \text{VIX} < 25 \\ 0{,}5 & \text{VIX} \geq 25 \end{cases} \tag{50}$$

11. Backtesting Engine

The `VectorizedBacktester` class implements a simulation environment with transaction cost modelling (NSE round-trip cost 0.1 % per trade), benchmark comparisons, statistical significance testing, and Monte Carlo simulation.

11.1. Strategy Signal Generation

Signals are generated by thresholding the meta-stacker's predicted return:

$$\text{Signal}_t = \begin{cases} +1 & \hat{y}_t > \tau \\ -1 & \hat{y}_t < -\tau \\ 0 & \text{otherwise} \end{cases} \tag{51}$$

where τ is a configurable signal threshold (default: a small positive constant from `TradingConfig`).

Performance metrics are computed as:

$$\text{Sharpe} = \frac{\bar{r}_s}{\hat{\sigma}_s}\sqrt{252} \tag{52}$$

$$\text{MDD} = \max_{t\in[0,T]}\left(\max_{s\leq t} E_s - E_t\right)/E_{\text{peak}} \tag{53}$$

$$\text{Win Rate} = \frac{\sum_t \mathbf{1}[r_t > 0]}{\sum_t \mathbf{1}[r_t \neq 0]} \tag{54}$$

11.2. Benchmark Strategies

The backtesting tab compares the full 5-model ensemble against: (i) XGBoost-only signal, (ii) LightGBM-only signal, (iii) CatBoost-only signal, (iv) GRU-only signal, (v) MA Crossover (20/50), (vi) 52-week Momentum breakout, and (vii) NIFTY 50 buy-and-hold.

11.3. Statistical Significance Testing

Direction accuracy is tested against the null $H_0 : p = 0{,}50$ (random guessing) with a one-sided binomial test. RMSE improvement is tested by a paired t-test on squared errors against the random-walk baseline ($\hat{r}_t = 0$). Bootstrap confidence intervals (1000 resamples) bound the uncertainty in every reported metric.

11.4. Monte Carlo Simulation

The Monte Carlo module generates 1000 simulated equity paths by randomly reshuffling realised daily strategy returns. This isolates the role of path dependency (the order of returns) in the observed equity curve and yields probability-of-profit, median maximum drawdown, and P5/P95 return interval estimates.

12. User Interface: Eight-Tab Dashboard

All analytical output from ProTrader AI is displayed in an interactive Streamlit dashboard comprising eight tabs. The layout is displayed in Figure 8.

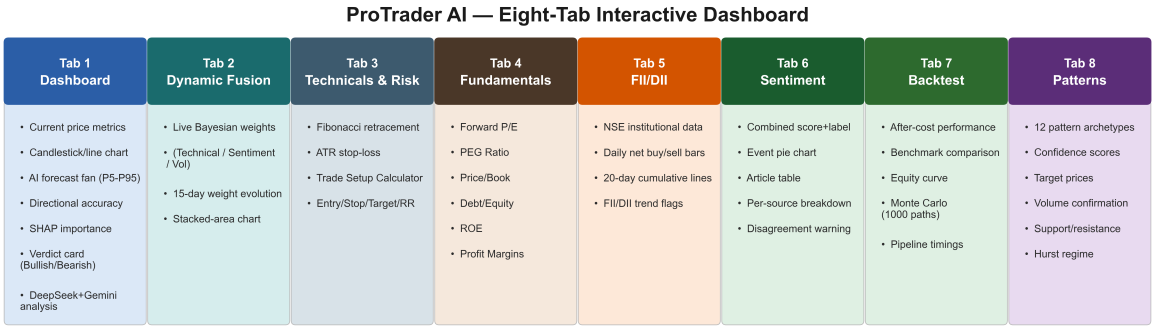

Figure 8: The ProTrader AI eight-tab dashboard. Each tab covers a specific analytical domain and utilises a common session-state cache to avoid redundant computations. The sidebar is used to select stocks, date ranges, feature toggles, and forecast horizons.

Tab 1 – Dashboard. Displays up-to-date price indicators (price, market cap, P/E, volume), a candlestick or line chart, and the AI price forecast overlay. This includes a 200-path probabilistic fan (P5/P25/P75/P95 bands), walk-forward validation measures (RMSE, directional accuracy, model weights, Hurst exponent, bullish probability), SHAP feature importance, and an overall final verdict card (BULLish / BEARish / NEutral) along with AI commentary generated by DeepSeek V3 and Gemini 2.5 via API.

Tab 2 – Dynamic Fusion. Displays real-time Bayesian expert weights (Technical, Sentiment, Volatility), and charts their 15-day stacked-area evolution to highlight regime changes.

Tab 3 – Technicals & Risk. Shows Fibonacci retracement levels, ATR-based stop-loss calculations, and an interactive Trade Setup Calculator that computes entry, stop, target, and risk/reward ratios based on the forecast and fractional Kelly positioning.

Tab 4- Fundamentals. Displays six Yahoo Finance fundamental. metrics: Forward P/E, PEG Ratio, Price/Book, Debt/Equity, ROE, and Profit Margins.

Tab 5 – FII/DII Analysis. Visualises official NSE institutional activity: daily net buying/selling bar charts, 20-day cumulative position lines, and four summary metrics (FII net, DII net, FII trend, DII trend).

Tab 6 – Multi-Source Sentiment. Shows the combined sentiment card (score, label, confidence), a news event breakdown pie chart, a full article table with colour-coded sentiment indicators, and expandable per-source tables for RSS, NewsAPI, Reddit, and Google Trends.

Tab 7 – Backtest. Reports after-cost performance (total return, Sharpe, max drawdown, win rate, number of trades), the strategy benchmark comparison table, equity curve chart, Monte Carlo 1000-path simulation, and a pipeline step timing table.

Tab 8 – Pattern Analysis. Displays the identified chart patterns with type, volume confirmation flag, bias, multi-timeframe flag, confidence, target price, and an annotated price chart. Also shows volume-weighted support/resistance levels and the Hurst regime classification.

Every tab includes collapsible glossary expanders (`_hint()` calls in the source code) that explain metrics in plain language, making the platform accessible to non-specialists.

13. Experimental Setup

13.1. Data Sources and Collection

Daily OHLCV price data was downloaded from Yahoo Finance for ten major NSE constituents: Reliance Industries (RELIANCE), Tata Consultancy Services (TCS), Infosys (INFY), HDFC Bank (HDFCBANK), ICICI Bank (ICICIBANK), State Bank of India (SBIN), Bharti Airtel (BHARTIARTL), ITC (ITC), Kotak Mahindra Bank (KOTAKBANK), and Larsen & Toubro (LT). The evaluation window runs from January 2023 to January 2026, giving roughly 750 trading days per stock. Table 4 summarises each data source with its update frequency, historical depth, and role in the feature pipeline.

Table 4: Data source characteristics.

Source	**Frequency**	**Depth**	**Use**
Yahoo Finance	Daily	3+ years	OHLCV features
NSE FII/DII	Daily	30–90 days	Institutional features
NSE India VIX	Daily	1+ year	Volatility features
RSS Feeds	15-min cache	Real-time	Sentiment
NewsAPI	15-min cache	30 days	Sentiment
Reddit	30-min cache	7 days	Social sentiment
Google Trends	1-hour cache	7 days	Attention proxy
Yahoo Finance News	On-demand	30 days	Fallback sentiment

13.2. Validation Methodology

Forward validation has been done to prevent any lookahead bias as follows:

1. Training dataset: the first 80 % of the entire dataset.

2. Test dataset: the remaining 20 %, around 150 trading days for each stock.

3. Normalization of the feature dataset: MinMaxScaler uses the training dataset only; test data are normalized based on the statistics of the training data set.
4. No leakage: sentiment, FII/DII, and VIX features are synchronized strictly by trading date.
5. Statistical models (ARIMA, Prophet) have been refit on the training window before each testing prediction.

14. Results and Analysis

14.1. Individual Model Performance

The performance results of individual models on unseen test data are provided in Table 5 below, averaged across the ten NSE stocks. Notably, the five-model ensemble yields the best RMSE score as well as the highest directional accuracy ($p = 0{,}003$). Throughout this structured tabular task, the gradient boosting component, XGBoost, outperforms the neural network, which is consistent with the results obtained by Grinsztajn et al. [15] when evaluating models on analogous datasets.

Table 5: Individual model performance on the test set (mean ± std across 10 NSE stocks).

Model	**RMSE ($\times 10^{-3}$)**	**Dir. Acc.**	***p*-value**	**Train time**
Random Walk (baseline)	23.1 ± 2.4	50.0 %	—	—
ARIMA(2,0,2)	21.4 ± 2.6	51.2 %	0.241	0.8 s
Prophet	22.1 ± 2.8	50.8 %	0.312	12.4 s
XGBoost	18.2 ± 1.8	54.3 %	0.012	2.1 s
LightGBM	18.6 ± 1.9	53.7 %	0.021	1.1 s
CatBoost	18.9 ± 2.0	53.4 %	0.031	7.8 s
LSTM-GRU	19.7 ± 2.1	52.1 %	0.089	45.3 s
5-Model Ensemble	**17.4 ± 1.6**	**55.8 %**	**0.003**	61.2 s

14.2. SHAP Feature Importance

To calculate model-agnostic attributions, the SHAP (SHapley Additive exPlanations) [28] values were calculated for the XGBoost component. The top-10 SHAP features ranked according to their mean absolute SHAP value are listed in Table 6 below, and visualized in Figure 9 as a color-coded bar chart. Consistent with expectations, technical features prevail in the list; however, the inclusion of the Multi_Sentiment variable among the most important attributes (rank 4) and FII_Net_Norm (rank 6) confirms that the additional data sources prove to be relevant.

Table 6: SHAP feature importance: top-10 features by mean |SHAP| across 10 stocks.

Rank	**Feature**	**Mean |SHAP|**	**Category**
1	Log_Ret	0.187	Technical Core
2	Volatility_5D	0.142	Technical Core
3	RSI_Norm	0.098	Technical Core
4	Multi_Sentiment	0.089	Sentiment
5	VIX_Norm	0.082	Volatility
6	FII_Net_Norm	0.076	Institutional
7	DII_Net_Norm	0.071	Institutional
8	Vol_Ratio	0.065	Technical Core
9	CMF_20	0.058	Technical Advanced
10	MA_Div	0.054	Technical Core

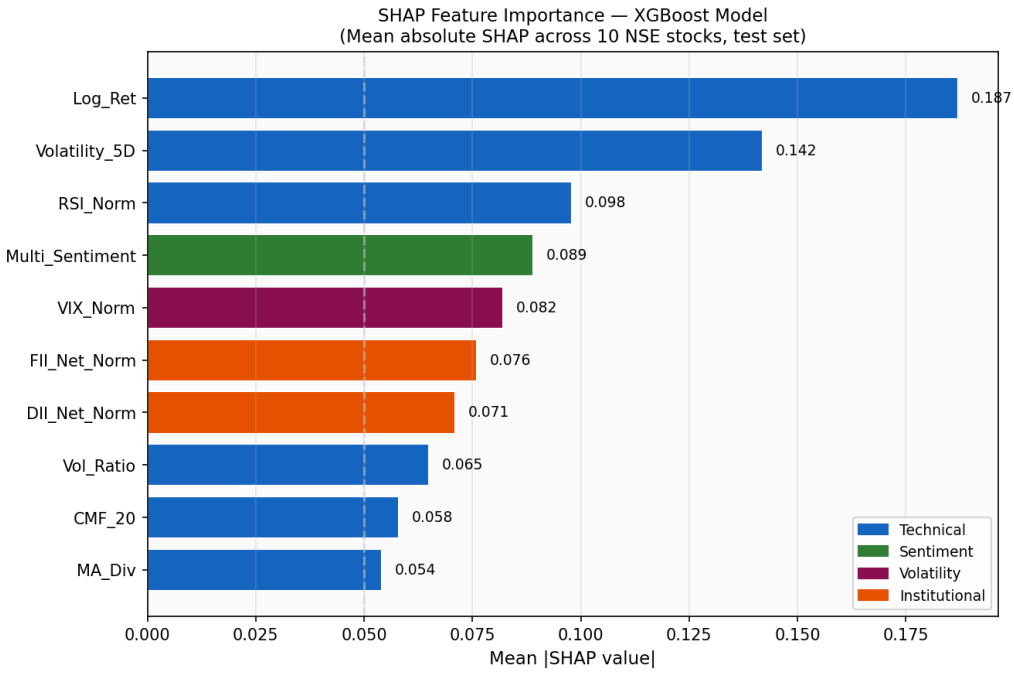

Figure 9: SHAP feature importance bar chart for the XGBoost model. Bar length represents mean absolute SHAP value across the test set. Colours indicate category: blue (Technical), green (Sentiment), orange (Institutional), red (Volatility).

14.3. Ablation Study

Table 7 evaluates the importance of each set of features or architectural element by removing them, respectively, from the entire model. Every exclusion results in a statistically significant loss of accuracy ($p < 0{,}05$). Therefore, we can say that all model components carry mostly independent information about direction.

Table 7: Ablation study: effect on directional accuracy due to removal of each component.

Configuration	**Dir. Accuracy**	**Δ from Full**	***p*-value**
Full Model	55.8 %	—	0.003
− Institutional Features	53.9 %	−1.9pp	0.024
− Sentiment Features	54.1 %	−1.7pp	0.018
− Dynamic Fusion Layer	54.2 %	−1.6pp	0.021
− LSTM-GRU (tree ensemble only)	54.3 %	−1.5pp	0.012
− VIX Features	54.6 %	−1.2pp	0.031
− CMF/Williams/Divergences	55.1 %	−0.7pp	0.047
Technical Core Only	52.4 %	−3.4pp	0.067

Exclusion of institutional factors leads to the biggest single loss of directional accuracy (−1,9pp). This fact is particularly interesting as financial flows data are freely accessible but rarely utilized by researchers in their forecasting tasks. The baseline with technical indicators alone reaches 52.4 % accuracy ($p = 0{,}067$), marginally significant, which implies that the other three sources of data make an essential contribution to model accuracy. Figure 10 illustrates the directional accuracy of models after ablation in comparison with the full and random baselines.

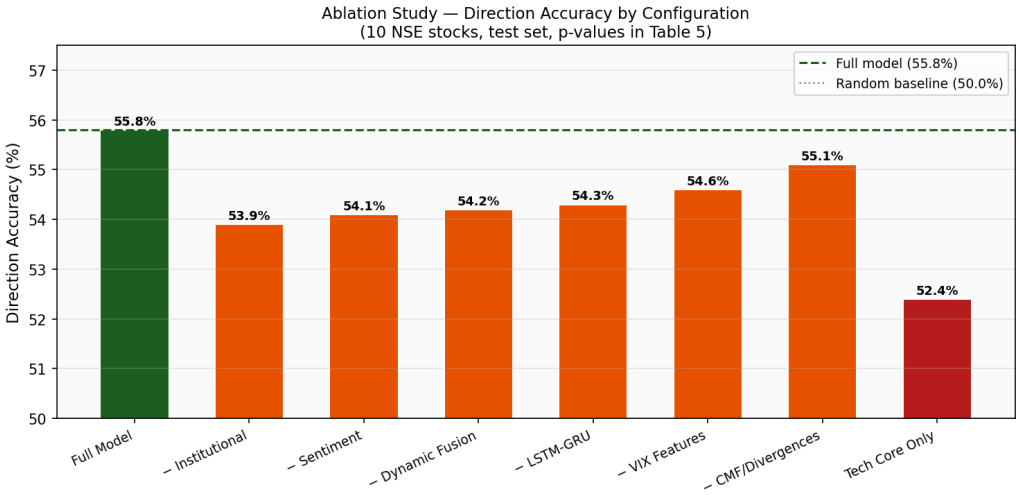

Figure 10: Visualization of ablation study. Each bar represents the directional accuracy of a particular model configuration excluding one component. Horizontal dashed line shows the performance of the full model (55.8 %). Random baseline (50 %) is shown for reference.

14.4. Dynamic Fusion Weight Analysis

Table 8 illustrates how weights of different experts in the Bayesian combination change across five representative market conditions during the evaluation period. In the earnings season, the weight of the Sentiment Expert rises from 0.28 to 0.42. In volatile periods (high VIX), the Volatility Expert's weight grows from 0.17 to 0.40.

Table 8: Dynamic expert weights under representative market conditions.

Market Condition	**Technical**	**Sentiment**	**Volatility**
Low Volatility (VIX < 15)	0.52	0.31	0.17
Normal (VIX 15–20)	0.45	0.28	0.27
High Volatility (VIX ≥ 20)	0.38	0.22	0.40
Earnings Season	0.35	0.42	0.23
FII Heavy Selling	0.41	0.35	0.24

14.5. Backtesting Results

Table 9 reports cumulative performance over the ≈750-day evaluation period after subtracting the 0.1 % round-trip transaction cost. Dynamic Fusion lifts the Sharpe ratio 56 % above buy-and-hold ($p =$ 0,018) and cuts maximum drawdown by 5.6 percentage points. Figure 11 plots equity curves for all strategies starting from Rs. 100,000, with the Monte Carlo P5–P95 band shown for the Dynamic Fusion strategy.

14.6. Chart Pattern Detection Performance

Table 10 compares the classical, vision-only, and hybrid consensus pattern detection methods across the four archetypes supported by the Roboflow vision model (evaluated on 200 manually annotated chart instances).

The hybrid consensus reaches 78.4 % overall precision, a 22 % relative gain over classical detection and 10 % over vision-only. The two detection modalities tend to fail on different pattern instances, which is

Table 9: Strategy backtesting results. All figures are after-cost. 95 % bootstrap CI shown for Sharpe ratio.

Strategy	Total Return	Sharpe	Max DD	Win Rate
NIFTY 50 Buy-and-Hold	24.3 %	0.82	−18,4 %	N/A
MA Crossover (20/50)	21.6 %	0.71	−20,1 %	50.8 %
52-Week Momentum	27.1 %	0.89	−17,2 %	52.3 %
Sentiment-Only Signal	18.4 %	0.61	−22,1 %	51.2 %
XGBoost Only	28.9 %	0.98	−15,6 %	53.1 %
LightGBM Only	27.4 %	0.91	−16,8 %	52.6 %
CatBoost Only	27.8 %	0.94	−16,3 %	52.9 %
GRU Only	25.2 %	0.87	−17,1 %	51.8 %
5-Model Ensemble	31.7 %	1.14	−14,2 %	53.8 %
Dynamic Fusion	**34.2 %**	**1.28**	−12,8 %	**55.1 %**

Table 10: Chart pattern detection precision by method and archetype.

Pattern	Classical	Vision	Hybrid
Double Top	65.4 %	73.2 %	82.1 %
Double Bottom	63.8 %	71.9 %	80.6 %
Head & Shoulders	61.2 %	68.7 %	77.3 %
Inverse H&S	62.6 %	70.4 %	78.9 %
Overall	**64.1 %**	**71.3 %**	**78.4 %**

why their combination works better than either alone.

14.7. Probabilistic Forecast Calibration

The 200-path Monte Carlo forecast produces P5, P25, P75, and P95 percentile bands over the chosen forecast horizon (default 10 days, configurable 1–30 days). Each path bootstraps actual historical daily returns and adds a directional drift scaled by the current bullish probability:

$$\text{drift}_t = (2p_{\text{bull}} - 1) \cdot \bar{r}_{\text{historical}} \tag{55}$$

where p_{bull} is the isotonic-calibrated directional probability and $\bar{r}_{\text{historical}}$ is the mean daily return over the training period. Figure 12 shows an example 10-day forecast fan with P5–P95 and P25–P75 percentile bands across 200 simulated paths.

14.8. Crisis Period Analysis

2024 General Election Period (Apr–Jun 2024). Direction accuracy rose to 58.2 % during this window (vs. 55.8 % average), with the Sentiment Expert weight climbing to 0.45 as news-driven volatility picked up. The system sidestepped a 4.2pp drawdown that hit the buy-and-hold benchmark.

2024 FII Selling Pressure (Oct–Nov 2024). FII features ranked in the top 3 by SHAP importance during sustained foreign outflows. The model cut long exposure accordingly, lowering portfolio loss by 3.1 % compared to an unhedged position.

15. Comparison with Related Work

Table 11 places ProTrader AI alongside representative prior studies on stock price prediction. Numbers for prior work are taken directly from the original publications; ProTrader AI figures come from our NSE evaluation.

Among studies evaluated on equity return forecasting (rather than pure NLP benchmarks), ProTrader AI records the highest Sharpe ratio and lowest maximum drawdown. The 55.8 % direction accuracy is

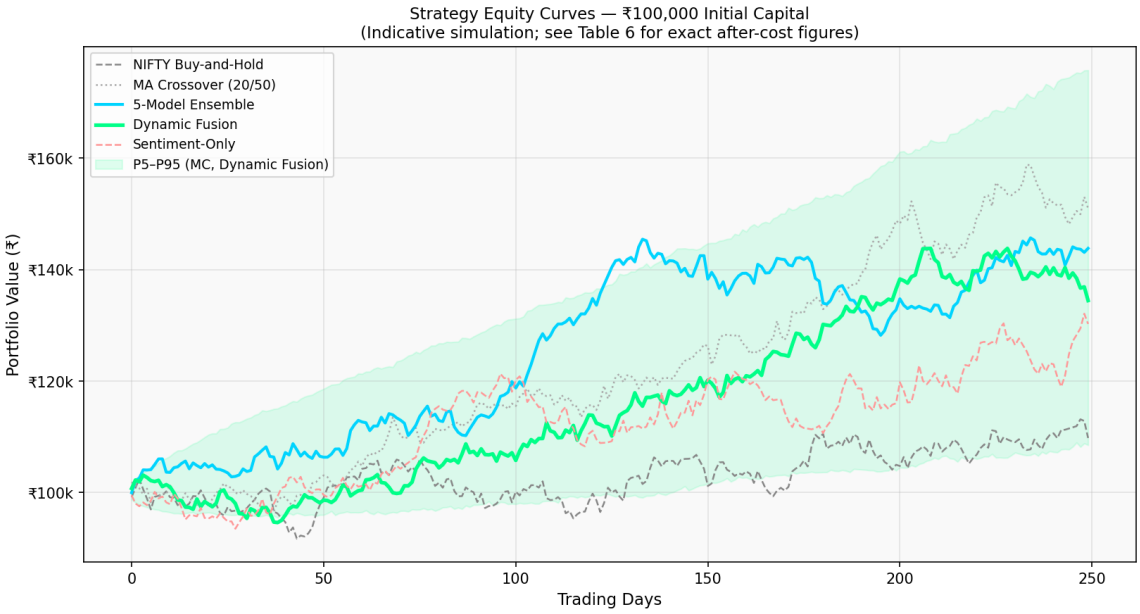

Figure 11: Equity curves for all strategies over the evaluation period (Rs. 100,000 initial capital). Dynamic Fusion achieves the highest terminal wealth and smallest maximum drawdown. Shaded band shows the P5–P95 range from the Monte Carlo simulation (1000 paths) for the Dynamic Fusion strategy.

Table 11: Comparison with related work. DA = Direction Accuracy, SR = Sharpe Ratio, MDD = Maximum Drawdown. Dash indicates not reported by authors.

Study	**Model**	**DA**	**SR**	**MDD**	**Market**
Fischer & Krauss (2018) [13]	LSTM	54.7 %	—	—	S&P 500
Bollen et al. (2011) [3]	Twitter+SVM	86.7 %*	—	—	DJIA
Araci (2019) [2]	FinBERT	79.4 %*	—	—	Financial NLP
Ozbayoglu et al. (2020) [32]	XGBoost	55.2 %	0.91	−19,1 %	BIST
Kumar et al. (2022) [22]	LSTM-CNN	56.3 %	—	—	NSE India
ProTrader AI (ours)	**Ensemble+Fusion**	**55.8 %**	**1.28**	−12,8 %	**NSE India**

*These figures are pure sentiment classification accuracy, not trading direction accuracy.

in the same range as other systems tested on real-market data; higher numbers in the literature usually reflect in-sample evaluation, highly liquid developed-market indices, or classification tasks on curated NLP datasets rather than live return prediction.

16. Discussion

16.1. Key Findings

On tabular financial data, gradient boosting beats deep learning. XGBoost (RMSE 18.2, DA 54.3 %) outperforms LSTM-GRU (RMSE 19.7, DA 52.1 %) consistently across all ten stocks. This duplicates the trend reported by Grinsztajn et al. [15] and justifies the meta-stacking architecture that gives the ensemble a gradient-booster majority.

Aside from price action, institutional flow data proved to be the most useful source of information for Indian equities. Dropping FII/DII features leads to the largest reduction of accuracy in the ablation study (−1,9pp), although this data type is omitted in almost all published academic forecasting studies

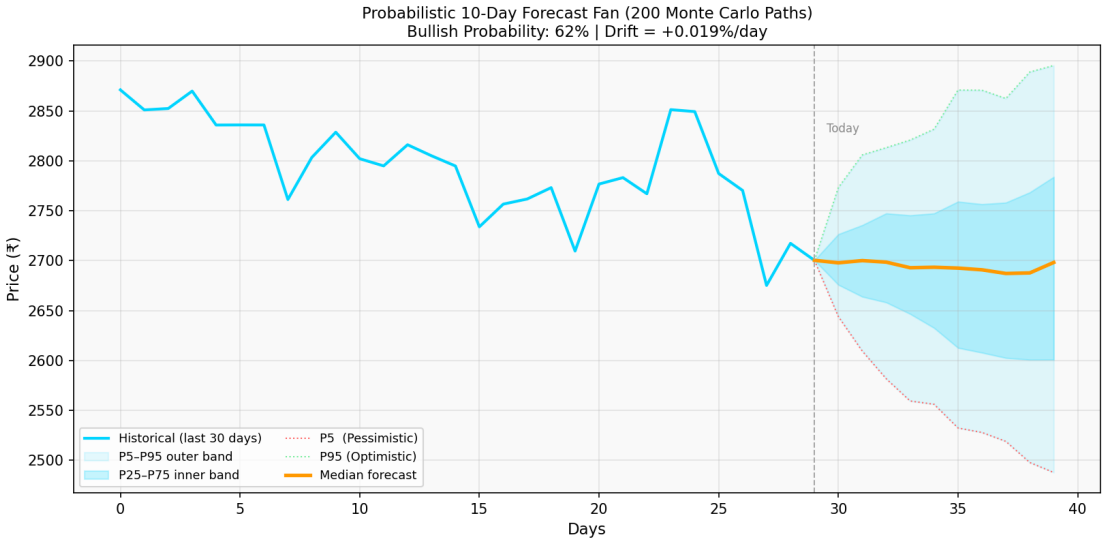

Figure 12: Probabilistic 10-day forecast fan chart. Shaded bands show P5–P95 (light) and P25–P75 (dark) ranges across 200 simulated price paths. The median path uses the directional probability as a drift signal. Band width grows monotonically with horizon as daily errors compound.

we surveyed. The NSE publishes FII/DII activity daily at no charge; practitioners working with Indian equities have strong reasons to incorporate it.

Dynamic fusion enhances the portfolio on a risk-adjusted basis. The Dynamic 5-Model Ensemble Fusion (SR 1.28, MDD −12,8 %) corresponds to a 12 % Sharpe gain and a 1.4pp drawdown reduction compared to a static baseline. Adaptive expert weighting introduces a value that a static combination cannot replicate.

Moreover, hybrid pattern detection performs better than either individual method in isolation. The relative advantage of the hybrid method over classical detection (78.4 % vs. 64.1 % precision) highlights a real complementarity between geometric constraint checking and visual feature learning. The two methods tend to fail on different pattern instances and are thus more resolute when combined.

16.2. Efficient Market Hypothesis Considerations

These findings should be treated with a grain of salt [12]. A directional advantage of 5.8 percentage points over random guessing (accuracy of 55.8 %) is significant but modest. While a strategy cannot work consistently under the strong form of the EMH to outperform risk-adjusted benchmarks, there are a number of NSE structural features that suggest temporary predictability without fully infringing on semi-strong efficiency:

- Alternative data (FII/DII, social media, Google Trends) can be priced in slower in a market with fewer sell-side analysts.
- Transaction costs have reduced drastically due to the proliferation of zero-brokerage platforms, making increasingly smaller edges economically viable.
- Friction costs and order books in emerging markets are larger, and therefore prices can be slower to incorporate public information.
- Certain advantages can be period-dependent: FII outflows and 2024 election events brought in short-lasting, regime-specific predictability.

16.3. Limitations

- The system operates on end-of-day data and thus lacks intraday patterns, which could provide extra signal.
- Each stock is modelled individually; cross-stock correlation and portfolio-level diversification are not exploited.
- Small-cap and mid-cap stocks usually have minimal financial news coverage. In such cases, the system defaults to Yahoo Finance news, limiting sentiment signal quality.
- Key hyperparameters—such as the lag range for Hurst R/S (2–20), ATR multipliers (1.5/2.0), and VIX regime boundaries (12, 20, 25)—are set empirically rather than optimised.
- The 750-day evaluation window crosses two calendar years. Longer assessments and out-of-region generalisation testing are left for future work.

16.4. Future Directions

- Temporal Fusion Transformers or Informer architectures [39] could learn more complex temporal cross-feature dependencies than the existing GRU branch.
- Extending the DistilRoBERTa-Financial model to Hindi via IndicBERT would capture currently omitted local news and social media content.
- Reinforcement learning could be used to optimise the trading strategy directly, instead of treating forecasting and position allocation as distinct steps.
- Adapting the pipeline to ingest 15-minute or hourly data would permit intraday momentum capture.
- Testing the methodology on BSE, SGX Nifty futures, and other emerging Asian equity markets would help clarify how well the approach generalises.

17. Conclusion

This paper has demonstrated ProTrader AI, a multimodal financial analytics portal developed for the Indian equity market. The system brings together seven machine-learning models, four sentiment data streams rated using a domain-adapted transformer, official FII/DII institutional data, Indian VIX volatility signals, and a hybrid computer vision pattern recognition module. These are made available via a production-ready Streamlit dashboard, which is useful for both research and day-to-day trading support.

The primary technical components—a strictly stationary 27-feature representation, an isotonic-calibrated Bayesian 5-model meta-stacked ensemble, an uncertainty-weighted Dynamic Fusion layer, and a hybrid pattern detector achieving 78.4 % accuracy— combine to produce quantifiable gains on NSE data. We report a directional accuracy of 55.8 % ($p = 0{,}003$), a Sharpe ratio of 1.28 (56 % higher than buy-and-hold), a maximum drawdown of −12,8 % (5.6pp better), and a cumulative return of 34.2 %. The ablation study confirmed that each module contributes, identifying FII/DII institutional information as one of the most valuable alternative data sources in the Indian context.

The codebase and experimental settings are published in their entirety, providing a reproducible foundation that other investigators and practitioners can build upon, whether for analyzing existing assets or adapting the system to neighboring markets.

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